Aaron Jacobs

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Education

Carnegie Mellon University Pittsburgh, PA
Master of Science in Computer Vision *Dec. 2027*
- Selected Coursework: Advanced Computer Vision, Visual Learning & Recognition, Robot Learning

Florida State University Tallahassee, FL
Bachelor of Science in Computer Engineering *May 2026*
Minor in Mathematics GPA: 3.93/4.00
- Relevant Coursework: Applied Machine Learning, Computer Vision, Deep and Reinforcement Learning

Skills

Languages: Python, C++, C, Java, Assembly, VHDL
ML/CV: PyTorch, TensorFlow, scikit-learn, OpenCV
Robotics/Systems: ROS 2, Linux, Docker, CMake
Data: NumPy, pandas, Matplotlib, MySQL
Tools: Git, Jupyter Notebook, AWS
Methods: SLAM, Depth Estimation, Sensor Fusion

Projects

Autonomous Racing Go-Kart | Florida State University - Senior Design Aug. 2025 – May 2026
- Architected the end-to-end autonomy software stack for a 5-person multidisciplinary racing team, deploying containerized ROS 2 on a Jetson Orin Nano and validating the system on an F1Tenth vehicle that completed autonomous laps with a 50 Hz closed-loop control pipeline.
- Integrated ZED 2i visual-inertial SLAM, classical line detection, and neural depth estimates to provide real-time localization, mapping, and track geometry under constrained edge-compute resources.
- Mitigated long-range depth-estimation variance by fitting a distance-calibration curve and selectively mapping higher-confidence near-field measurements, improving geometric consistency for downstream planning and control.
- Designed a modular compute architecture separating high-level perception and planning from low-level vehicle actuation, enabling the autonomy stack to be developed independently of the full-scale go-kart hardware.

NASA Sponsored ML Competition Nov. 2024 – Dec. 2024
- Developed an LSTM pipeline to predict U.S. airport arrival throughput 3 hours ahead in 15-minute intervals, achieving a 2.8 flight prediction error standard deviation versus 9.8 flights for the baseline, a 71% reduction.
- Processed and feature-engineered 200+ GB of airport configuration, weather, and flight-log data using Python, transforming heterogeneous raw records into leakage safe temporal training data.
- Delivered the complete data processing, training, and inference pipeline in a 2-person team over a 2-week sprint.

Autonomous Drone May 2024 – Dec. 2024
- Designed and built a custom autonomous drone from the ground up, integrating a 3D-printed airframe, motors, ESCs, microcontroller, onboard vision, and autonomous navigation into a complete robotic system.
- Developed a lightweight multi-scale circular-filter detector for ceiling light localization, processing onboard camera frames in real time and feeding continuously updated target estimates into a 100 Hz flight-control loop.
- Designed target-tracking and navigation logic to persist and update position estimates through intermittent visual detections, enabling stable, repeatable autonomous approaches to detected bulbs across end-to-end flight trials.

Research

Undergraduate Researcher Jan. 2025 – Jan. 2026
Adversarial Machine Learning, Zhixin Pan *Florida State University*
- Designed and executed approximately 10 controlled experiments evaluating whether knowledge distillation could mitigate backdoors in CNNs trained on poisoned image datasets.
- Implemented data-poisoning and distillation pipelines and evaluated self-distillation and teacher-ensemble distillation using clean accuracy and attack success rate (ASR).
- Reduced backdoor ASR from approximately 90% to 32% through distillation while retaining 84% clean accuracy vs 91% pre-distillation, demonstrating a measurable tradeoff between Trojan suppression and predictive performance.

Experience

Moffitt Cancer Center May 2023 – Aug. 2023
Radiation Therapy Assistant
- Worked alongside Radiation Therapists to support treatment workflows and deliver high-quality patient care.
- Maintained efficiency and situational awareness in a fast-paced, high-stress hospital environment.
- ”I don’t have the words to tell you just how grateful I am for making a scary situation a lot less scary. Your encouraging words, attention to details, and kindness has blessed not just my life but the lives of my family members and my praying friends. I can’t say thank you enough!” -Patient Thank You Card